

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402528
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Kim; Mikyung et al.

Organic light-emitting device and apparatus including the same

Abstract

An organic light-emitting device includes: a first electrode; a second electrode facing the first electrode; an organic layer disposed between the first electrode and the second electrode and includes an emission layer; and a first compound and a second compound; wherein the first compound includes a hole transport compound not including an electron transport moiety, and the second compound includes an electron transport compound including an electron transport moiety or a bipolar compound including an electron transport moiety, wherein the second compound includes deuterium.

Inventors: Kim; Mikyung (Yongin-si, KR), Lee; Jaehyun (Yongin-si, KR), Yang; Seunggak (Yongin-si, KR), Park; Byounghee (Yongin-si, KR), Seo; Jihyun (Yongin-si, KR)

Applicant: Samsung Display Co., LTD. (Yongin-si, KR)

Family ID: 1000008779289

Assignee: Samsung Display Co., Ltd. (Yongin-si, KR)

Appl. No.: 17/060532

Filed: October 01, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210104698 A1	Apr. 08, 2021

Foreign Application Priority Data

KR	10-2019-0123357	Oct. 04, 2019
----	-----------------	---------------

Publication Classification

Int. Cl.: H10K85/60 (20230101); H10K50/11 (20230101); H10K101/00 (20230101);
H10K101/10 (20230101); H10K102/00 (20230101)

U.S. Cl.:

CPC H10K85/654 (20230201); H10K85/6572 (20230201); H10K50/11 (20230201);
H10K2101/10 (20230201); H10K2101/90 (20230201); H10K2102/351 (20230201)

Field of Classification Search

CPC: H01L (51/0067); H01L (51/0072); H01L (2251/5384); H01L (51/0061); H10K (85/654);
H10K (85/6572); H10K (2101/90)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6677060	12/2003	Li et al.	N/A	N/A
9093650	12/2014	Kim et al.	N/A	N/A
9893290	12/2017	Min et al.	N/A	N/A
10050212	12/2017	Cho et al.	N/A	N/A
10326111	12/2018	Yoo et al.	N/A	N/A
2010/0084967	12/2009	Takeda	313/504	H01L 51/0072
2011/0101312	12/2010	LeCloux	544/212	H01L 51/0508
2011/0147718	12/2010	Howard, Jr.	N/A	N/A
2013/0248849	12/2012	Feldman et al.	N/A	N/A
2013/0264561	12/2012	Dobbs et al.	N/A	N/A
2014/0336392	12/2013	Kim et al.	N/A	N/A
2015/0236262	12/2014	Cho	257/40	H01L 51/0054
2015/0380662	12/2014	Kim	257/40	H10K 85/6572
2016/0260905	12/2015	Lee	N/A	H01L 51/006
2016/0372683	12/2015	Tanimoto	N/A	C09K 11/06
2017/0018600	12/2016	Ito et al.	N/A	N/A
2017/0025618	12/2016	Zheng et al.	N/A	N/A
2017/0047527	12/2016	Lee	N/A	N/A
2017/0084845	12/2016	Kim et al.	N/A	N/A
2017/0092873	12/2016	Kang	N/A	N/A
2017/0162796	12/2016	Jeong	N/A	C07D 241/48
2017/0200899	12/2016	Kim et al.	N/A	N/A
2018/0138425	12/2017	Ma et al.	N/A	N/A
2021/0395215	12/2020	Kim	N/A	C09K 11/06

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10-2014-0015298	12/2013	KR	N/A
10-2014-0015299	12/2013	KR	N/A
10-2014-0080451	12/2013	KR	N/A
10-2015-0124902	12/2014	KR	N/A

10-2016-0003362	12/2015	KR	N/A
10-2016-0107406	12/2015	KR	N/A
20160122889	12/2015	KR	N/A
10-2017-0037277	12/2016	KR	N/A
10-1934385	12/2018	KR	N/A
2010071362	12/2009	WO	N/A

OTHER PUBLICATIONS

Hedley, Gordon J. et al “Ultrafast luminescence in Ir(ppy)₃.” Chemical Physical Letters. 450. 4-6(2008): 292-296 (Year: 2008). cited by examiner

Nakamura et al., The hydrogen/deuterium isotope effect of the host material on the lifetime of organic light-emitting diodes, Chem. Commun., 2014, 50, pp. 14870-14872. cited by applicant

Primary Examiner: Chandhok; Jenna N

Attorney, Agent or Firm: Womble Bond Dickinson (US) LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority from and the benefit of Korean Patent Application No. 10-2019-0123357, filed on Oct. 4, 2019, which is hereby incorporated by reference for all purposes as if fully set forth herein.

BACKGROUND

Field

(2) Exemplary implementations of the invention relate generally to an organic light-emitting device and, more specifically, to an apparatus including the same.

Discussion of the Background

(3) Organic light-emitting devices are self emissive devices and, compared with a device of the related art, they have a wide viewing angle, a high contrast ratio, and a short response time, and show excellent characteristics in terms of luminance, driving voltage, and response speed.

(4) In an organic light-emitting device, a first electrode is arranged on a substrate, and a hole transport region, an emission layer, an electron transport region, and a second electrode are sequentially formed on the first electrode. Holes provided from the first electrode may move toward the emission layer through the hole transport region, and electrons provided from the second electrode may move toward the emission layer through the electron transport region. The holes and the electrons, which are carriers, recombine in the emission layer to produce excitons. These excitons transition from an excited state to a ground state, thereby generating light. However, many existing organic light-emitting devices suffer from reduced lifespans during operation.

(5) The above information disclosed in this Background section is only for understanding of the background of the inventive concepts, and, therefore, it may contain information that does not constitute prior art.

SUMMARY

(6) Applicant discovered that by combining emission layers of organic light-emitting devices with different attributes unexpected synergistic improvements in efficiency and lifespan are obtained.

(7) Organic emitting-light devices and apparatus including the same constructed according to the principles and exemplary implementations of the invention have high efficiency and long lifespan. For example, the emission layer of an organic light-emitting device composed of one or more

layers of certain compounds that that satisfy certain of the equations disclosed herein, the emission layer has reduced resistance. In addition, the organic light-emitting device appropriately maintains ratios of holes and electrons based on the difference of the highest occupied molecular orbital HOMO energy levels of the hosts in the emission layer. For example, the organic light-emitting device may have first and second emission layers that satisfy certain of the disclosed equations such that the first emission layer has long lifespan, but relatively low efficiency, and the second emission layer has high efficiency, but relatively short lifespan, but the overall efficiency and lifespan of the organic light-emitting device are improved.

(8) Additional features of the inventive concepts will be set forth in the description which follows, and in part will be apparent from the description, or may be learned by practice of the inventive concepts.

(9) According to one aspect of the invention, an organic light-emitting device includes: a first electrode; a second electrode facing the first electrode; an organic layer disposed between the first electrode and the second electrode and includes an emission layer; and a first compound and a second compound; wherein the first compound includes a hole transport compound not including an electron transport moiety, and the second compound includes an electron transport compound including an electron transport moiety or a bipolar compound including an electron transport moiety, wherein the second compound includes deuterium.

(10) The electron transport moiety may include at least one of a cyano group, a phosphine oxide group, a sulfone oxide group, a sulfonate group, and a π electron-depleted nitrogen-containing ring.

(11) The first compound may be a compound of Formulae 301-1 to 301-5, or a compound of HT1-1, HT1-2, HT2-1, or HT2-2, as disclosed herein.

(12) The first compound may be a fluorene-containing compound, a carbazole-containing compound, a diarylamine compound, a triarylamine compound, a dibenzofuran-containing compound, a dibenzothiophene-containing compound, or a dibenzosilole-containing compound.

(13) The first compound may include deuterium.

(14) The second compound may be a compound of Formulae 302-1 to 302-4, or a compound of ET2-1, ET2-2, ET4-1, or ET4-2, as disclosed herein.

(15) The second compound may be a pyridine-containing compound, a pyrimidine-containing compound, a triazine-containing compound, a triazole-containing compound, an imidazole-containing compound, or an oxazine-containing compound.

(16) The second compound may be a compound of Formula 303 or of Formulae 303-1 to 303-6, as disclosed herein.

(17) The emission layer may include the first compound and the second compound.

(18) The emission layer may include a first emission layer and a second emission layer, the first emission layer may include a first host and a second host, and the second emission layer may include a third host and a fourth host, wherein, the first host in the first emission layer and the third host in the second emission layer may each include the first compound, and the first host and the third host are same or different from each other, the second host in the first emission layer and the fourth host in the second emission layer may each include the second compound, the second compound included in the second host may include the electron transport compound, the second compound included in the fourth host may include the bipolar compound, the first host and the second host may each satisfy Equation 1 disclosed herein, and the third host and the fourth host may each satisfy Equation 2 disclosed herein.

(19) The emission layer may emit green light having a maximum emission wavelength in a range of about 490 nm to about 560 nm.

(20) The emission layer may further include a dopant, and the dopant may include a phosphorescent dopant or a fluorescent dopant.

(21) The first electrode may be an anode, the second electrode may be a cathode, the organic layer may further include a hole transport region disposed between the first electrode and the emission

layer, and an electron transport region disposed between the emission layer and the second electrode; the hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and the electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or any combination thereof.

(22) The hole transport region may include an amine-based compound.

(23) The electron transport region may include a metal-containing material.

(24) An apparatus may include an organic light-emitting device, wherein the first electrode of the organic light-emitting device may be in electrical contact with one of a source electrode and a drain electrode of a thin-film transistor.

(25) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are intended to provide further explanation of the invention as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification, illustrate exemplary embodiments of the invention, and together with the description serve to explain the inventive concepts.

(2) FIG. 1 is a schematic cross-sectional diagram of an exemplary embodiment of an organic light-emitting device constructed according to principles of the invention.

(3) FIG. 2 is a schematic cross-sectional diagram of another exemplary embodiment of an organic light-emitting device constructed according to principles of the invention.

(4) FIG. 3 is a schematic cross-sectional diagram of yet another exemplary embodiment of an organic light-emitting device constructed according to principles of the invention.

(5) FIG. 4 is a schematic cross-sectional diagram of still another exemplary embodiment of an organic light-emitting device constructed according to principles of the invention.

(6) FIG. 5 is a schematic diagram of an exemplary embodiment of an apparatus containing an organic light-emitting device constructed according to principles of the invention.

DETAILED DESCRIPTION

(7) In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of various exemplary embodiments or implementations of the invention. As used herein “embodiments” and “implementations” are interchangeable words that are non-limiting examples of devices or methods employing one or more of the inventive concepts disclosed herein. It is apparent, however, that various exemplary embodiments may be practiced without these specific details or with one or more equivalent arrangements. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring various exemplary embodiments. Further, various exemplary embodiments may be different, but do not have to be exclusive. For example, specific shapes, configurations, and characteristics of an exemplary embodiment may be used or implemented in another exemplary embodiment without departing from the inventive concepts.

(8) Unless otherwise specified, the illustrated exemplary embodiments are to be understood as providing exemplary features of varying detail of some ways in which the inventive concepts may be implemented in practice. Therefore, unless otherwise specified, the features, components, modules, layers, films, panels, regions, and/or aspects, etc. (hereinafter individually or collectively referred to as “elements”), of the various embodiments may be otherwise combined, separated, interchanged, and/or rearranged without departing from the inventive concepts.

(9) The use of cross-hatching and/or shading in the accompanying drawings is generally provided to clarify boundaries between adjacent elements. As such, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, dimensions, proportions, commonalities between illustrated elements, and/or any other characteristic, attribute, property, etc., of the elements, unless specified. Further, in the accompanying drawings, the size and relative sizes of elements may be exaggerated for clarity and/or descriptive purposes. When an exemplary embodiment may be implemented differently, a specific process order may be performed differently from the described order. For example, two consecutively described processes may be performed substantially at the same time or performed in an order opposite to the described order. Also, like reference numerals denote like elements.

(10) When an element, such as a layer, is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it may be directly on, connected to, or coupled to the other element or layer or intervening elements or layers may be present. When, however, an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. To this end, the term “connected” may refer to physical, electrical, and/or fluid connection, with or without intervening elements. Further, the D1-axis, the D2-axis, and the D3-axis are not limited to three axes of a rectangular coordinate system, such as the x, y, and z-axes, and may be interpreted in a broader sense. For example, the D1-axis, the D2-axis, and the D3-axis may be perpendicular to one another, or may represent different directions that are not perpendicular to one another. For the purposes of this disclosure, “at least one of X, Y, and Z” and “at least one selected from the group consisting of X, Y, and Z” may be construed as X only, Y only, Z only, or any combination of two or more of X, Y, and Z, such as, for instance, XYZ, XYY, YZ, and ZZ. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(11) Although the terms “first,” “second,” etc. may be used herein to describe various types of elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another element. Thus, a first element discussed below could be termed a second element without departing from the teachings of the disclosure.

(12) Spatially relative terms, such as “beneath,” “below,” “under,” “lower,” “above,” “upper,” “over,” “higher,” “side” (e.g., as in “sidewall”), and the like, may be used herein for descriptive purposes, and, thereby, to describe one elements relationship to another element(s) as illustrated in the drawings. Spatially relative terms are intended to encompass different orientations of an apparatus in use, operation, and/or manufacture in addition to the orientation depicted in the drawings. For example, if the apparatus in the drawings is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. Furthermore, the apparatus may be otherwise oriented (e.g., rotated 90 degrees or at other orientations), and, as such, the spatially relative descriptors used herein interpreted accordingly.

(13) The terminology used herein is for the purpose of describing particular embodiments and is not intended to be limiting. As used herein, the singular forms, “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Moreover, the terms “comprises,” “comprising,” “includes,” and/or “including,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components, and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. It is also noted that, as used herein, the terms “substantially,” “about,” and other similar terms, are used as terms of approximation and not as terms of degree, and, as such, are utilized to account for inherent deviations in measured, calculated, and/or provided values that would be recognized by one of ordinary skill in the art.

(14) Unless otherwise defined, all terms (including technical and scientific terms) used herein have

the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure is a part. Terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an idealized or overly formal sense, unless expressly so defined herein.

(15) An aspect of the invention provides an organic light-emitting device including: a first electrode; a second electrode facing the first electrode; an organic layer disposed between the first electrode and the second electrode and including an emission layer; a first compound; and a second compound.

(16) The first compound may be a hole transport compound not including an electron transport moiety, and the second compound may be an electron transport compound including an electron transport moiety or a bipolar compound including an electron transport moiety, wherein the second compound may include deuterium.

(17) In one exemplary embodiment, the electron transport moiety may include at least one selected from a cyano group, a phosphine oxide group, a sulfone oxide group, a sulfonate group, and a π electron-depleted nitrogen-containing ring.

(18) In one exemplary embodiment, the electron transport moiety may include at least one selected from: a cyano group, a phosphine oxide group, a sulfone oxide group, and a sulfonate group; a C.sub.1-C.sub.60 alkyl group, a phenyl group, a naphthyl group, a biphenyl group, and a terphenyl group, each substituted with at least one selected from a cyano group, a phosphine oxide group, a sulfone oxide group, and a sulfonate group; and a group represented by one of Formulae 7-1 to 7-59 below:

(19) ##STR00001## ##STR00002## ##STR00003## ##STR00004## ##STR00005##
##STR00006##

(20) In Formulae 7-1 to 7-59, Z.sub.51 and Z.sub.52 may each independently be at least one selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, an imidazolyl group, a pyrazole group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, pyrimidinyl, a pyridazinyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a thiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; g₂ may be an integer from 0 to 2, g₃ may be an integer from 0 to 3 integer, g₄ may be an integer from 0 to 4 integer, g₅ may be an integer from 0 to 5 integer, g₆ may be an integer from 0 to 6, and * and *' each indicate a binding site to a neighboring site.

(21) In one exemplary embodiment, the electron transport moiety may include at least one selected from a pyridine group, a pyrimidine group, a triazine group, a quinoline group, an isoquinoline group, and a quinazoline group.

(22) In one exemplary embodiment, the first compound may be a compound represented by one of Formulae 301-1 to 301-5 below:

(23) ##STR00007##

(24) In Formulae 301-1 to 301-5, A.sub.301 to A.sub.304 may each independently be selected from a benzene group, a naphthalene group, a phenanthrene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, an indole group, a carbazole group, a benzocarbazole group, a dibenzocarbazole group, a furan group, a benzofuran group, a

dibenzofuran group, a naphthofuran group, a benzonaphthofuran group, a dinaphthofuran group, a thiophene group, a benzothiophene group, a dibenzothiophene group, a naphthothiophene group, a benzonaphthothiophene group, and a dinaphthothiophene group, X.sub.301 may be O, S, N-[(L.sub.305).sub.xb5-R.sub.305], C[(L.sub.305).sub.xb5-R.sub.305][(L.sub.306).sub.xb6-R.sub.306], or Si[(L.sub.305).sub.xb5-R.sub.305][(L.sub.306).sub.xb6-R.sub.306], X.sub.302, Y.sub.301, and Y.sub.302 may each independently be a single bond, O, S, N-[(L.sub.305).sub.xb5-R.sub.305], C[(L.sub.305).sub.xb5-R.sub.305][(L.sub.306).sub.xb6-R.sub.306], Si[(L.sub.305).sub.xb5-R.sub.305][(L.sub.306).sub.xb6-R.sub.306], or S(=O).sub.2, xb22 and xb23 may each independently be 0, 1, or 2, L.sub.301 to L.sub.307 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, xb1 to xb6 may each independently be an integer from 0 to 5, xb7 may be an integer from 1 to 5, R.sub.301 to R.sub.306 may each independently be selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.301)(Q.sub.302)(Q.sub.303), —N(Q.sub.301)(Q.sub.302), —B(Q.sub.301)(Q.sub.302), —C(=O)(Q.sub.301), —S(=O).sub.2(Q.sub.301), and —P(=O)(Q.sub.301)(Q.sub.302), R.sub.311 to R.sub.314 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group —Si(Q.sub.301)(Q.sub.302)(Q.sub.303), —N(Q.sub.301)(Q.sub.302), —B(Q.sub.301)(Q.sub.302), —C(=O)(Q.sub.301), —S(=O).sub.2(Q.sub.301), and —P(=O)(Q.sub.301)(Q.sub.302), and Q.sub.301 to Q.sub.303 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group.

(25) In one exemplary embodiment, the first compound may be selected from a fluorene-containing compound, a carbazole-containing compound, a diarylamine compound, a triarylamine compound, a dibenzofuran-containing compound, a dibenzothiophene-containing compound, and a dibenzosilole-containing compound.

(26) In one exemplary embodiment, the first compound may include deuterium.

(27) In one exemplary embodiment, in Formulae 301-1 to 301-5, at least one selected from R.sub.301 to R.sub.306 and R.sub.311 to R.sub.314 may be deuterium, a C.sub.1-C.sub.60 alkyl group substituted with deuterium, a C.sub.3-C.sub.10 cycloalkyl group substituted with deuterium, a C.sub.6-C.sub.60 aryl group substituted with deuterium, or a C.sub.1-C.sub.60 heteroaryl group substituted with deuterium.

(28) In one exemplary embodiment, A.sub.301 to A.sub.304 in Formulae 301-1 to 301-5 may each independently be selected from a benzene group, an indene group, a naphthalene group, an anthracene group, a fluorene group, a phenanthrene group, and a carbazole group.

(29) In one exemplary embodiment, A.sub.31, A.sub.32, and A.sub.34 in Formulae 301-1 to 301-5 may each independently be selected from a benzene group, a naphthalene group, and a phenanthrene group.

(30) In one exemplary embodiment, A.sub.31, A.sub.32, and A.sub.34 in Formulae 301-1 to 301-5 may each independently be a benzene group or a naphthalene group.

(31) In one exemplary embodiment, L.sub.301 to L.sub.307 in Formulae 301-1 to 301-5 may each independently be selected from:

(32) a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group, each substituted with at least one selected from deuterium, —F, —Cl,

—Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32).

(33) In one exemplary embodiment, R.sub.301 to R.sub.307 in Formulae 301-1 to 301-5 may each independently be selected from: a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, and a dibenzosilolyl group; and a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, and a dibenzosilolyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, and a dibenzosilolyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32).

(34) In one exemplary embodiment, Q.sub.31 to Q.sub.33 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(35) In one exemplary embodiment, the second compound may be an electron transport compound

including an electron transport moiety.

(36) In one or more exemplary embodiments, the second compound may be a compound represented by one of Formulae 302-1 to 302-4 below:

(37) ##STR00008##

(38) In Formulae 302-1 to 302-4, X.sub.321 to X.sub.327 may each independently be N or C[(L.sub.324).sub.xb24-R.sub.324], Y.sub.31 may be O or S, X.sub.328 may be N[(L.sub.325).sub.xb25-R.sub.325] or C[(L.sub.325).sub.xb25-R.sub.325][(L.sub.326).sub.xb26-R.sub.326], xb21 to xb26 may each independently be an integer from 0 to 5, L.sub.321 to L.sub.326 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, R.sub.321 to R.sub.326 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.321)(Q.sub.322)(Q.sub.323), —N(Q.sub.321)(Q.sub.322), —B(Q.sub.321)(Q.sub.322), —C(=O)(Q.sub.321), —S(=O).sub.2(Q.sub.321), and —P(=O)(Q.sub.321)(Q.sub.322), at least one selected from R.sub.321 to R.sub.326 may include deuterium, two or more substituents selected from R.sub.321 to R.sub.326 may optionally be linked to each other to form a substituted or unsubstituted C.sub.1-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, and Q.sub.321 to Q.sub.323 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group.

(39) In one exemplary embodiment, at least one selected from R.sub.321 to R.sub.326 in Formulae 302-1 to 302-4 may be deuterium, a C.sub.1-C.sub.60 alkyl group substituted with deuterium, a C.sub.3-C.sub.10 cycloalkyl group substituted with deuterium, a C.sub.6-C.sub.60 aryl group substituted with deuterium, or a C.sub.1-C.sub.60 heteroaryl group substituted with deuterium.

(40) In one exemplary embodiment, the second compound may be a pyridine-containing compound, a pyrimidine-containing compound, a triazine-containing compound, a triazole-containing compound, an imidazole-containing compound, or an oxazine-containing compound.

(41) In one exemplary embodiment, the second compound may be a bipolar compound including

an electron transport moiety.

(42) In one exemplary embodiment, the second compound may be a compound represented by Formula 303 below:

(43) ##STR00009##

(44) In Formula 303, A.sub.31, A.sub.32, and A.sub.34 may each independently be selected from a C.sub.5-C.sub.60 carbocyclic group and a C.sub.1-C.sub.30 heterocyclic group, A.sub.33 may be a group represented by Formula 303A, X.sub.31 may be selected from N[(L.sub.335).sub.xb35-(R.sub.335)], O, S, Se, C[(L.sub.335).sub.xb35-(R.sub.335)][(L.sub.336).sub.xb36-(R.sub.336)], and Si[(L.sub.335).sub.xb35-(R.sub.335)][(L.sub.336).sub.xb36-(R.sub.336)], xb31 to xb36 may each independently be an integer from 0 to 5, L.sub.331 to L.sub.336 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, R.sub.331 to R.sub.336 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.331)(Q.sub.332)(Q.sub.333), —N(Q.sub.331)(Q.sub.332), —B(Q.sub.331)(Q.sub.332), —C(=O)(Q.sub.331), —S(=O).sub.2(Q.sub.331), and —P(=O)(Q.sub.331)(Q.sub.332), at least one selected from L.sub.331 to L.sub.336 and R.sub.331 to R.sub.336 may include an electron transport moiety, at least one selected from R.sub.331 to R.sub.336 may include deuterium, xb42 to xb44 may each independently be an integer from 0 to 10, and Q.sub.331 to Q.sub.333 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group.

(45) In one exemplary embodiment, at least one selected from R.sub.331 to R.sub.336 in Formula 303 may be deuterium, a C.sub.1-C.sub.60 alkyl group substituted with deuterium, a C.sub.3-C.sub.10 cycloalkyl group substituted with deuterium, a C.sub.6-C.sub.60 aryl group, the substituted with deuterium, or a C.sub.1-C.sub.60 heteroaryl group substituted with deuterium.

(46) In one exemplary embodiment, the second compound may be a compound represented by one of Formulae 303-1 to 303-6 below:

(47) ##STR00010##

(48) In Formulae 303-1 to 303-6, A.sub.31, A.sub.34, X.sub.31, xb31 to xb34, L.sub.331 to L.sub.336, R.sub.331 to R.sub.336, xb42, and xb44 may each be understood by referring to the corresponding descriptions thereof provided above, and xb43 may be 1 or 2.

(49) In one exemplary embodiment, A.sub.31, A.sub.32, and A.sub.34 in Formula 303 may each independently be selected from a benzene group, an indene group, a naphthalene group, an anthracene group, a fluorene group, a phenanthrene group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, and a carbazole group.

(50) In one exemplary embodiment, A.sub.31, A.sub.32, and A.sub.34 in Formula 303 may each independently be selected from a benzene group, a naphthalene group, a phenanthrene group, a pyridine group, a pyrimidine group, a pyrazine group, and a pyridazine group.

(51) In one exemplary embodiment, A.sub.31, A.sub.32, and A.sub.34 in Formula 303 may each independently be a benzene group, a naphthalene group, or a phenanthrene group.

(52) In one exemplary embodiment, A.sub.31, A.sub.32, and A.sub.34 in Formula 303 may each independently be a benzene group or a naphthalene group.

(53) In one exemplary embodiment, L.sub.321 to L.sub.326 and L.sub.331 to L.sub.336 in Formulae 302-1 to 302-4 and 303 may each independently be selected from: a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group, each substituted with at least one selected from deuterium, —F, —Cl,

—Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32).

(54) In one exemplary embodiment, R.sub.321 to R.sub.326 and R.sub.331 to R.sub.336 in Formulae 302-1 to 302-4 and 303 may each independently be selected from: a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group; and a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl

group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), and —N(Q.sub.31)(Q.sub.32).

(55) In one exemplary embodiment, Q.sub.31 to Q.sub.33 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(56) In one exemplary embodiment, Q.sub.301 to Q.sub.303, Q.sub.321 to Q.sub.323, and Q.sub.331 to Q.sub.333 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group.

(57) In one exemplary embodiment, the first compound may be selected from Compounds HT1-1, HT1-2, HT2-1, and HT2-2 below:

(58) ##STR00011## ##STR00012##

(59) In one exemplary embodiment, the first compound may be selected from Compounds HT2-1 and HT2-2 below:

(60) In one exemplary embodiment, the second compound may be selected from Compounds ET2-1, ET2-2, ET4-1, and ET4-2 below:

(61) ##STR00013##

(62) The organic light-emitting device may include the first compound and the second compound. The first compound may be a hole transport compound not including an electron transport moiety, and the second compound may be an electron transport compound including an electron transport moiety or a bipolar compound including an electron transport moiety, wherein a combination of the first compound and the second compound may control the distribution of holes and electrons in the organic light-emitting device.

(63) The second compound may include an electron transport moiety and deuterium. Deuterium is an isotope of hydrogen, and a deuterium-carbon bond thereof is more stable than a hydrogen-carbon bond, and thus, the second compound may have high molecular stability, and more particularly, may have excellent electrical stability.

(64) Therefore, according to one exemplary embodiment, when the emission layer of the organic light-emitting device is formed by mixing or co-depositing the first compound and the second compound, injection characteristics of holes and electrons are improved, and charge balance in the emission layer is also improved, thereby increasing both efficiency and lifespan.

(65) In one exemplary embodiment, when the first compound includes deuterium, the molecular stability, especially the electrical stability, of the first compound may be improved, and accordingly, the organic light-emitting device including the first compound and the second compound may have further improved efficiency and further improved lifespan.

(66) In one exemplary embodiment, the first electrode may be an anode.

(67) In one exemplary embodiment, the second electrode may be a cathode.

(68) In one exemplary embodiment, the organic light-emitting device may further include a hole transport region between the first electrode and the emission layer and an electron transport region

between the emission layer and the second electrode.

(69) In one exemplary embodiment, the hole transport region may include at least one selected from a hole injection layer, a hole transport layer, a buffer layer, an emission auxiliary layer, and an electron blocking layer.

(70) In one exemplary embodiment, the electron transport region may include at least one selected from a hole blocking layer, an electron transport layer, and an electron injection layer.

(71) In one exemplary embodiment, the hole transport region may include an amine-based compound.

(72) In one exemplary embodiment, the first compound and the second compound may be included in the emission layer.

(73) In one exemplary embodiment, based on weights of the first compound and the second compound, an amount of the first compound in the emission layer may be greater than that of the second compound.

(74) In one exemplary embodiment, the emission layer may emit green light having the maximum emission wavelength in a range of about 490 nm to about 560 nm.

(75) In one exemplary embodiment, the emission layer may further include a dopant, and the dopant may include a phosphorescent dopant or a fluorescent dopant.

(76) In one exemplary embodiment, the emission layer may further include a dopant, and the dopant may include a phosphorescent dopant.

(77) In one exemplary embodiment, the electron transport region may include a metal-containing material.

(78) In one exemplary embodiment, the first emission layer may further include a first dopant, and the second emission layer may further include a second dopant. In one exemplary embodiment, the first dopant and the second dopant may be identical to or different from each other. The first dopant and the second dopant may each be understood by referring to the descriptions provided in connection with the dopant.

(79) Another aspect of the invention provides an apparatus including: the organic light-emitting device; and a thin-film transistor, wherein the first electrode of the organic light-emitting device is in electrical contact with one of a source electrode or a drain electrode of the thin-film transistor.

(80) The term “organic layer” as used herein refers to a single layer and/or all layers between the first electrode and the second electrode of the organic light-emitting device. A material included in “the organic layer” is not limited to an organic material.

(81) Description of FIG. 1

(82) FIG. 1 is a schematic view of an organic light-emitting device **10** according to one exemplary embodiment. The organic light-emitting device **10** includes a first electrode **110**, an organic layer **150**, and a second electrode **190**.

(83) Hereinafter, the structure of the organic light-emitting device **10** according to an embodiment and a method of manufacturing the organic light-emitting device **10** will be described in connection with FIG. 1.

(84) First Electrode **110**

(85) Referring to FIG. 1, a substrate may be additionally located under the first electrode **110** or above the second electrode **190**. The substrate may be a glass substrate or a plastic substrate, each having excellent mechanical strength, thermal stability, transparency, surface smoothness, ease of handling, and water resistance.

(86) The first electrode **110** may be, for example, formed by depositing or sputtering a material for forming the first electrode **110** on the substrate. When the first electrode **110** is an anode, the material for a first electrode **110** may be selected from materials with a high work function to facilitate hole injection.

(87) The first electrode **110** may be a reflective electrode, a semi-transmissive electrode, or a transmissive electrode. When the first electrode **110** is a transmissive electrode, the material for

forming the first electrode **110** may be selected from indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO₂), zinc oxide (ZnO), and any combinations thereof, but exemplary embodiments of the invention are not limited thereto. In one or more exemplary embodiments, when the first electrode **110** is a semi-transmissive electrode or a reflective electrode, the material for forming the first electrode **110** may be selected from magnesium (Mg), silver (Ag), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), and any combinations thereof, but exemplary embodiments of the invention are not limited thereto.

(88) The first electrode **110** may have a single-layered structure, or a multi-layered structure including two or more layers. For example, the first electrode **110** may have a three-layered structure of ITO/Ag/ITO, but the structure of the first electrode **110** is not limited thereto.

(89) Organic Layer **150**

(90) The organic layer **150** is disposed on the first electrode **110**. The organic layer **150** may include an emission layer.

(91) The organic layer **150** may further include a hole transport region between the first electrode **110** and the emission layer and an electron transport region between the emission layer and the second electrode **190**.

(92) Hole Transport Region in Organic Layer **150**

(93) The hole transport region may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

(94) The hole transport region may include at least one layer selected from a hole injection layer, a hole transport layer, an emission auxiliary layer, and an electron blocking layer.

(95) For example, the hole transport region may have a single-layered structure including a single layer including a plurality of different materials, or a multi-layered structure having a hole injection layer/hole transport layer structure, a hole injection layer/hole transport layer/emission auxiliary layer structure, a hole injection layer/emission auxiliary layer structure, a hole transport layer/emission auxiliary layer structure, or a hole injection layer/hole transport layer/electron blocking layer structure, wherein for each structure, constituting layers are sequentially stacked from the first electrode **110** in this stated order, but the structure of the hole transport region is not limited thereto.

(96) In one exemplary embodiment, the hole transport region may include at least one selected from 4,4',4''-Tris[phenyl(m-tolyl)amino]triphenylamine (m-MTDATA), 1-N,1-N-bis[4-(diphenylamino)phenyl]-4-N,4-N-diphenylbenzene-1,4-diamine (TDATA), 4,4',4''-Tris[2-naphthyl(phenyl)amino]triphenylamine (2-TNATA), N,N'-Di(1-naphthyl)-N,N'-diphenyl-(1,1'-biphenyl)-4,4'-diamine (NPB or NPD), N₄N₄'-Di(naphthalen-2-yl)-N₄N₄'-diphenyl-[1,1'-biphenyl]-4,4'-diamine (β-NPB), N,N'-Bis(3-methylphenyl)-N,N'-diphenylbenzidine (TPD), 4(N,N'-Bis(3-methylphenyl)-N,N'-diphenyl-9,9-spirobifluorene-2,7-diamine (spiro-TPD), 2,7-Bis[N-(1-naphthyl)anilino]-9,9'-spirobi[9H-fluorene] (spiro-NPB), 2,2'-Dimethyl-N,N'-di-[(1-naphthyl)-N,N'-diphenyl]-1,1'-biphenyl-4,4'-diamine (methylated-NPB), 4-(1-{4-[bis(4-methylphenyl)amino]phenyl}cyclohexyl)-N,N-bis(4-methylphenyl)aniline (TAPC), N,N,N',N'-Tetrakis(3-methylphenyl)-3,3'-dimethylbenzidine (HMTDP), 4,4',4''-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), a compound represented by Formula 201 below, and a compound represented by Formula 202 below:

(97) ##STR00014## ##STR00015## ##STR00016##

(98) In Formulae 201 and 202, L.sub.201 to L.sub.204 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted

C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylenylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylenylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, L.sub.205 may be selected from *—O—* , *—S—* , *—N(Q.sub.201)—* , a substituted or unsubstituted C.sub.1-C.sub.20 alkylene group, a substituted or unsubstituted C.sub.2-C.sub.20 alkenylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylenylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylenylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, xa1 to xa4 may each independently be an integer from 0 to 3, xa5 may be an integer from 1 to 10, R.sub.201 to R.sub.204 and Q.sub.201 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

(99) In one exemplary embodiment, in Formula 202, R.sub.201 and R.sub.202 may optionally be linked via a single bond, a dimethyl-methylene group, or a diphenyl-methylene group, and R.sub.203 and R.sub.204 may optionally be linked via a single bond, a dimethyl-methylene group, or a diphenyl-methylene group.

(100) In one exemplary embodiment, in Formulae 201 and 202, L.sub.201 to L.sub.205 may each independently be selected from: a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylenylene group, a heptalenylene group, an indacenylene group, an acenaphthylenylene group, a fluorenylenylene group, a spiro-bifluorenylenylene group, a benzofluorenylenylene group, a dibenzofluorenylenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a thiophenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylene group, a benzothiophenylene group, a dibenzofuranylene group, a dibenzothiophenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group; and a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylenylene group, a heptalenylene group, an indacenylene group, an acenaphthylenylene group, a fluorenylenylene group, a spiro-bifluorenylenylene group, a benzofluorenylenylene group, a dibenzofluorenylenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a thiophenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylene group, a benzothiophenylene group, a dibenzofuranylene group, a dibenzothiophenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a

perylene group, a pentaphenyl group, a hexacene group, a pentacene group, a rubicene group, a coronene group, an ovalene group, a thiophene group, a furan group, a carbazole group, an indole group, an isoindole group, a benzofuran group, a benzothiophene group, a dibenzofuran group, a dibenzothiophene group, a benzocarbazole group, a dibenzocarbazole group, a dibenzosilole group, a pyridine group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33) and —N(Q.sub.31)(Q.sub.32); and Q.sub.31 to Q.sub.33 are the same as described above.

(104) In one or more exemplary embodiments, at least one selected from R.sub.201 to R.sub.203 in Formula 201 may each independently be selected from: a fluorene group, a spiro-bifluorene group, a carbazole group, a dibenzofuran group, and a dibenzothiophene group, a fluorene group, a spiro-bifluorene group, a carbazole group, a dibenzofuran group, and a dibenzothiophene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a naphthyl group, a fluorene group, a spiro-bifluorene group, a carbazole group, a dibenzofuran group, and a dibenzothiophene group, but exemplary embodiments of the invention are not limited thereto.

(105) In one or more exemplary embodiments, in Formula 202, i) R.sub.201 and R.sub.202 may be linked via a single bond, and/or ii) R.sub.203 and R.sub.204 may be linked via a single bond.

(106) In one or more exemplary embodiments, R.sub.201 to R.sub.204 in Formula 202 may be selected from; a carbazole group; and a carbazole group substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a naphthyl group, a fluorene group, a spiro-bifluorene group, a carbazole group, a dibenzofuran group, and a dibenzothiophene group, but exemplary embodiments of the invention are not limited thereto.

(107) In one or more exemplary embodiments, the compound represented by Formula 201 may be represented by Formula 201A below:

(108) ##STR00017##

(109) In one or more exemplary embodiments, the compound represented by Formula 201 may be represented by Formula 201A(1) below, but exemplary embodiments of the invention are not limited thereto:

(110) ##STR00018##

(111) In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201A-1 below, but exemplary embodiments of the invention are not limited thereto:

(112) ##STR00019##

(113) In one or more exemplary embodiments, the compound represented by Formula 202 may be represented by Formula 202A below:

(114) ##STR00020##

(115) In one or more exemplary embodiments, the compound represented by Formula 202 may be represented by Formula 202A-1 below:

(116) ##STR00021##

(117) In Formulae 201A, 201A(1), 201A-1, 202A, and 202A-1, L.sub.201 to L.sub.203, xa1 to xa3, xa5, and R.sub.202 to R.sub.204 may each be understood by referring to the corresponding descriptions thereof provided above, R.sub.211 and R.sub.212 may each be understood by referring to the description presented in connection with R.sub.203, and R.sub.213 to R.sub.217 may each

independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group.

(118) The hole transport region may include at least one compound selected from Compounds HT1 to HT39 below, but compounds to be included in the hole transport region are not limited thereto:

(119) ##STR00022## ##STR00023## ##STR00024## ##STR00025## ##STR00026##
##STR00027## ##STR00028## ##STR00029##

(120) The thickness of the hole transport region may be in a range of about 100 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å. When the hole transport region includes at least one of a hole injection layer and a hole transport layer, the thickness of the hole injection layer may be in a range of about 100 Å to about 9,000 Å, for example, about 100 Å to about 1,000 Å, and the thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å, for example about 100 Å to about 1,500 Å. When the thicknesses of the hole transport region, the hole injection layer and the hole transport layer are within these ranges, satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

(121) The emission auxiliary layer may increase light-emission efficiency by compensating for the optical resonance distance according to the wavelength of light emitted by an emission layer, and the electron blocking layer may block the flow of electrons from an electron transport region. The emission auxiliary layer and the electron blocking layer may include the materials as described above.

(122) p-Dopant

(123) The hole transport region may further include, in addition to these materials, a charge-generation material for the improvement of conductive properties. The charge-generation material may be homogeneously or non-homogeneously dispersed in the hole transport region.

(124) The charge-generation material may be, for example, a p-dopant.

(125) In one exemplary embodiment, a lowest unoccupied molecular orbital (LUMO) of the p-dopant may be -3.5 eV or less.

(126) The p-dopant may include at least one selected from a quinone derivative, a metal oxide, and a cyano group-containing compound, but exemplary embodiments of the invention are not limited thereto.

(127) For example, the p-dopant may include at least one selected from a quinone derivative, such as tetracyanoquinodimethane (TCNQ) and F4-2,3,5,6-Tetrafluoro-7,7,8,8-tetracyanoquinodimethane (TCNQ); a metal oxide, such as tungsten oxide or molybdenum oxide; 1,4,5,8,9,12-hexaazatriphenylene-hexacarbonitrile (HAT-CN); and a compound represented by Formula 221 below, but exemplary embodiments of the invention are not limited thereto:

(128) ##STR00030##

(129) In Formula 221, R.sub.221 to R.sub.223 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10

cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, wherein at least one selected from R.sub.221 to R.sub.223 may have at least one substituent selected from a cyano group, —F, —Cl, —Br, —I, a C.sub.1-C.sub.20 alkyl group substituted with —F, a C.sub.1-C.sub.20 alkyl group substituted with —Cl, a C.sub.1-C.sub.20 alkyl group substituted with —Br, and a C.sub.1-C.sub.20 alkyl group substituted with —I.

Emission Layer in Organic Layer **150**

(130) When the organic light-emitting device **10** is a full-color organic light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, or a blue emission layer, according to a representative sub-pixel. In one or more exemplary embodiments, the emission layer may have a stacked structure of two or more layers selected from a red emission layer, a green emission layer, and a blue emission layer, in which the two or more layers contact each other or are separated from each other. In one or more exemplary embodiments, the emission layer may include two or more materials selected from a red light-emitting material, a green light-emitting material, and a blue light-emitting material, in which the two or more materials are mixed with each other in a single layer to emit white light.

(131) The emission layer may include at least one host and a dopant. The dopant may include at least one selected from a phosphorescent dopant and a fluorescent dopant.

(132) In the emission layer, an amount of the dopant may be in a range of about 0.01 parts by weight to about 15 parts by weight based on 100 parts by weight of the host, but exemplary embodiments of the invention are not limited thereto.

(133) The thickness of the emission layer may be in a range of about 100 Å to about 1,000 Å, for example, about 200 Å to about 600 Å. When the thickness of the emission layer is within this range, excellent light-emission characteristics may be obtained without a substantial increase in driving voltage.

(134) At Least One Host in Emission Layer

(135) In one exemplary embodiment, the emission layer may include a first emission layer and a second emission layer, wherein the first emission layer includes a first host and a second host, and the second emission layer includes a third host and a fourth host.

(136) In one exemplary embodiment, in the first emission layer and the first host may be formed from the first compound, and the second host may be formed from the second compound. For example, in the first emission layer, the first host may be a hole transport compound not including the electron transport moiety, and the second host may be an electron transport compound including an electron transport moiety. In one or more exemplary embodiments, in the first emission layer, the first host may be a hole transport compound not including an electron transport moiety, and the second host may be a bipolar compound including an electron transport moiety.

(137) In one exemplary embodiment, in the second emission layer and the third host may be formed from the first compound, and the fourth host may be formed from the second compound. For example, in the second emission layer and the third host may be formed from a hole transport compound not including an electron transport moiety, and the fourth host may be formed from an electron transport compound including an electron transport moiety. In one or more exemplary embodiments, in the second emission layer and the third host may be formed from a hole transport compound not including an electron transport moiety, and the fourth host may be formed from a bipolar compound including an electron transport moiety.

(138) In one exemplary embodiment, the first host in the first emission layer and the third host in the second emission layer may each be understood by referring to the descriptions provided herein of the first compound.

(139) In one exemplary embodiment, the second host in the first emission layer and the fourth host in the second emission layer may each be understood by referring to the descriptions provided herein of the second compound.

(140) In one exemplary embodiment, the first host and the third host may be identical to or different from each other. In one exemplary embodiment, the second host and the fourth host may be identical to or different from each other.

(141) In one exemplary embodiment, the first host and the third host may be different from each other, and the second host and the fourth host may be different from each other.

(142) In one exemplary embodiment, the first host in the first emission layer and the third host in the second emission layer may each be selected from the first compound, and may be identical to or different from each other.

(143) In one exemplary embodiment, the second host in the first emission layer and the fourth host in the second emission layer may each be selected from the second compound, and may be identical to or different from each other.

(144) In one exemplary embodiment, the first host in the first emission layer and the third host in the second emission layer may each include the first compound, and may be identical to or different from each other,

(145) The second host in the first emission layer and the fourth host in the second emission layer may each include the second compound, the second compound in the second host may be the electron transport compound, and the second compound in the fourth host may be the bipolar compound.

(146) Referring now to the equations below, the first host and the second host may each satisfy Equation 1 below, and the third host and the fourth host may each satisfy Equation 2 below:

$$E_{\text{sub.HOMO1}} - E_{\text{sub.HOMO2}} \leq 0.2 \text{ eV} \quad \text{<Equation 1>}$$

$$E_{\text{sub.HOMO3}} - E_{\text{sub.HOMO4}} \geq 0.3 \text{ eV} \quad \text{<Equation 2>}$$

(147) In Equations 1 and 2, $E_{\text{sub.HOMO1}}$ indicates a highest occupied molecular orbital (HOMO) of the first host, $E_{\text{sub.HOMO2}}$ indicates a HOMO energy level of the second host, $E_{\text{sub.HOMO3}}$ indicates a HOMO energy level of the third host, and $E_{\text{sub.HOMO4}}$ indicates a HOMO energy level of the fourth host.

(148) When the organic light-emitting device satisfies Equations 1 and 2 the emission layer has reduced resistance. In addition, the organic light-emitting device satisfying Equations 1 and 2 appropriately maintains ratios of holes and electrons based on the difference of the HOMO energy levels of the hosts in the emission layer. That is, in the organic light-emitting device satisfying Equations 1 and 2, the first emission layer, which has long lifespan, but relatively low efficiency, and the second emission layer, which has high efficiency, but relatively short lifespan, are appropriately used, thereby improving efficiency and lifespan of the organic light-emitting device at the same time.

(149) In detail, when the first host and the second host in the first emission layer satisfy Equation 1, the first emission layer may have improved hole transport characteristics. Since the first emission layer has excellent hole transport characteristics, the hole transport from the first electrode toward the first emission layer may be smoothly performed. In further detail, the first host and the second host in the first emission layer may further satisfy Equation 1-1 below:

$$0 \text{ eV} \leq E_{\text{sub.HOMO1}} - E_{\text{sub.HOMO2}} \leq 0.2 \text{ eV} \quad \text{<Equation 1-1>}$$

(150) In addition, when the third host and the fourth host in the second emission layer satisfy Equation 1, the second emission layer may perform less hole transport than the first emission layer. Due to low performance of the hole transport in the second emission, holes in the emission layer do not form excitons, and accordingly, the movement of the excitons toward the second electrode may be presented, thereby increasing emission efficiency of the organic light-emitting device. In further detail, the third host and the fourth host in the second emission layer may further satisfy Equation 2-1 below:

0.3 eV ≤ E.sub.HOMO3 - E.sub.HOMO4 ≤ 0.8 eV <Equation 2-1>

(151) The at least one host may include the first compound and the second compound.

(152) The at least one host may further include a compound represented by Formula 301 below:

[Ar.sub.301].sub.xb11-[(L.sub.301).sub.xb1-R.sub.301].sub.xb21 <Formula 301>

(153) In Formula 301, A.sub.301 may be a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, xb11 may be 1, 2, or 3, L.sub.301 may be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, xb1 may be an integer from 0 to 5, R.sub.301 may be selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.301)(Q.sub.302)(Q.sub.303), —N(Q.sub.301)(Q.sub.302), —B(Q.sub.301)(Q.sub.302), —C(=O)(Q.sub.301), —S(=O).sub.2(Q.sub.301), and —P(=O)(Q.sub.301)(Q.sub.302), xb21 may be an integer from 1 to 5, and Q.sub.301 to Q.sub.303 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group, but exemplary embodiments of the invention are not limited thereto.

(154) In one exemplary embodiment, Ar.sub.301 in Formula 301 may be selected from: a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, and a dibenzothiophene group; and a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, and a dibenzothiophene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group, but exemplary embodiments of the invention are not limited thereto.

(155) When xb11 in Formula 301 is 2 or more, two or more Ar.sub.301(s) may be linked to each

other via a single bond.

(156) In one or more exemplary embodiments, the host may include an alkaline earth metal complex. For example, the host may include a Be complex (for example, Compound H55 below), a Mg complex, and a Zn complex.

(157) The at least one host may further include at least one selected from 9,10-di(2-naphthyl)anthracene (ADN), 2-methyl-9,10-bis(naphthalen-2-yl)anthracene (MADN), 9,10-di-(2-naphthyl)-2-t-butyl-anthracene (TBADN), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 1,3-di-9-carbazolylbenzene (mCP), 1,3,5-tri(carbazol-9-yl)benzene (TCP), and Compounds H1 to H55 below, but exemplary embodiments of the invention are not limited thereto:

(158) ##STR00031## ##STR00032## ##STR00033## ##STR00034## ##STR00035##
##STR00036## ##STR00037## ##STR00038## ##STR00039## ##STR00040## ##STR00041##
##STR00042## ##STR00043##

Phosphorescent Dopant Included in Emission Layer in Organic Layer 150

(159) The phosphorescent dopant may include an organometallic complex represented by Formula 401 below:

(160) ##STR00044##

(161) In Formulae 401 and 402, M may be selected from iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), rhodium (Rh), and thulium (Tm), L.sub.401 may be a ligand represented by Formula 402, and xc1 may be 1, 2, or 3, wherein when xc1 is two or more, two or more L.sub.401(s) may be identical to or different from each other, L.sub.402 may be an organic ligand, and xc2 may be an integer from 0 to 4, wherein when xc2 may be two or more, two or more L.sub.402(s) may be identical to or different from each other, X.sub.401 to X.sub.404 may each independently be nitrogen or carbon, X.sub.401 and X.sub.403 may be linked via a single bond or a double bond, and X.sub.402 and X.sub.404 may be linked via a single bond or a double bond, A.sub.401 and A.sub.402 may each independently be a C.sub.5-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, X.sub.405 may be a single bond, *—O—*', *—S—*', *—C(=O)—*', *—N(Q.sub.411)—*', *—C(Q.sub.411)(Q.sub.412)—*', *—C(Q.sub.411)=C(Q.sub.412)—*', *—C(Q.sub.411)=*', or *=C=*', wherein Q.sub.411 and Q.sub.412 may each independently be hydrogen, deuterium, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group, X.sub.406 may be a single bond, O, or S, R.sub.401 and R.sub.402 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.20 alkyl group, a substituted or unsubstituted C.sub.1-C.sub.20 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.401)(Q.sub.402)(Q.sub.403), —N(Q.sub.401)(Q.sub.402), —B(Q.sub.401)(Q.sub.402), —C(=O)(Q.sub.401), —S(=O).sub.2(Q.sub.401), and —P(=O)(Q.sub.401)(Q.sub.402), and Q.sub.401 to Q.sub.403 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a C.sub.6-C.sub.20 aryl group, and a C.sub.1-C.sub.20 heteroaryl group, xc11 and xc12 may each independently be an integer from 0 to 10, * and *' in Formula 402 each indicate a binding site to M in Formula 401.

(162) In one exemplary embodiment, A.sub.401 and A.sub.402 in Formula 402 may each independently be selected from a benzene group, a naphthalene group, a fluorene group, a spiro-

bifluorene group, an indene group, a pyrrole group, a thiophene group, a furan group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a quinoxaline group, a quinazoline group, a carbazole group, a benzimidazole group, a benzofuran group, a benzothiophene group, an isobenzothiophene group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a dibenzofuran group, and a dibenzothiophene group.

(163) In one or more exemplary embodiments, in Formula 402, i) X.sub.401 may be nitrogen and X.sub.402 may be carbon, or ii) X.sub.401 and X.sub.402 may each be nitrogen at the same time.

(164) In one or more exemplary embodiments, R.sub.401 and R.sub.402 in Formula 402 may each independently be selected from: hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, and a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group and a C.sub.1-C.sub.20 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a phenyl group, a naphthyl group, a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, and a norbornenyl group; a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group; a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, an adamantyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group; and —Si(Q.sub.401)(Q.sub.402) (Q.sub.403), —N(Q.sub.401)(Q.sub.402), —B(Q.sub.401)(Q.sub.402), —C(=O)(Q.sub.401), —S(=O).sub.2(Q.sub.401), and —P(=O)(Q.sub.401)(Q.sub.402), and Q.sub.401 to Q.sub.403 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, and a naphthyl group, but exemplary embodiments of the invention are not limited thereto.

(165) In one or more exemplary embodiments, when xc1 in Formula 401 is two or more, two A.sub.401(s) in two or more L.sub.401(s) may optionally be linked to each other via X.sub.407, which is a linking group, two A.sub.402(s) may optionally be linked to each other via X.sub.408, which is a linking group (see Compounds PD1 to PD4 and PD7 below). X.sub.407 and X.sub.408 may each independently be a single bond, *—O—*, *—S—*, *—C(=O)—*, *—N(Q.sub.413)—*, *—C(Q.sub.413)(Q.sub.414)—* or *—C(Q.sub.413)=C(Q.sub.414)—* (where Q.sub.413 and Q.sub.414 may each independently be hydrogen, deuterium, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group), but exemplary embodiments of the invention are not limited thereto.

(166) L.sub.402 in Formula 401 may be a monovalent, divalent, or trivalent organic ligand. For

example, L.sub.402 may be selected from halogen, diketone (for example, acetylacetonate), carboxylic acid (for example, picolinate), —C(=O) , isonitrile, —CN , and phosphorus (for example, phosphine or phosphite), but exemplary embodiments of the invention are not limited thereto.

(167) In one or more exemplary embodiments, the phosphorescent dopant may be selected from, for example, Compounds PD1 to PD25 below, but exemplary embodiments of the invention are not limited thereto:

(168) ##STR00045## ##STR00046##

Fluorescent Dopant in Emission Layer

(169) The fluorescent dopant may include an arylamine compound or a styrylamine compound.

(170) The fluorescent dopant may include a compound represented by Formula 501 below:

(171) ##STR00047##

(172) In Formula 501, Ar.sub.501 may be a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, L.sub.501 to L.sub.503 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, xd1 to xd3 may each independently be an integer from 0 to 3, R.sub.501 and R.sub.502 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, and xd4 may be an integer from 1 to 6.

(173) In one exemplary embodiment, Ar.sub.501 in Formula 501 may be selected from: a naphthalene group, a heptalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, and an indenophenanthrene group; and a naphthalene group, a heptalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, and an indenophenanthrene group, each substituted with at least one selected from deuterium, —F , —Cl , —Br , —I , a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group;

(174) In one or more exemplary embodiments, L.sub.501 to L.sub.503 in Formula 501 may each independently be selected from: a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a

hexacenylene group, a pentacenylene group, a thiophenylene group, a furanylene group, a carbazolyne group, an indolyne group, an isoindolyne group, a benzofuranylene group, a benzothiophenylene group, a dibenzofuranylene group, a dibenzothiophenylene group, a benzocarbazolyne group, a dibenzocarbazolyne group, a dibenzosilolyne group, a pyridinyne group; and a phenylene group, a naphthyne group, a fluorenyne group, a spiro-bifluorenyne group, a benzofluorenyne group, a dibenzofluorenyne group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a thiophenylene group, a furanylene group, a carbazolyne group, an indolyne group, an isoindolyne group, a benzofuranylene group, a benzothiophenylene group, a dibenzofuranylene group, a dibenzothiophenylene group, a benzocarbazolyne group, a dibenzocarbazolyne group, a dibenzosilolyne group, and a pyridinyne group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group.

(175) In one or more exemplary embodiments, R.sub.501 and R.sub.502 in Formula 501 may each independently be selected from: a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group; and a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, and —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), and Q.sub.31 to Q.sub.33 may each dependently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(176) In one or more exemplary embodiments, xd4 in Formula 501 may be 2, but exemplary embodiments of the invention are not limited thereto.

(177) For example, the fluorescent dopant may be selected from Compounds FD1 to FD22 below:

(178) ##STR00048## ##STR00049##

(179) In one or more exemplary embodiments, the fluorescent dopant may be selected from the following compounds, but exemplary embodiments of the invention are not limited thereto.

(180) ##STR00050##

Electron Transport Region in Organic Layer 150

(181) The electron transport region may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

(182) The electron transport region may include at least one selected from a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, and an electron injection layer, but exemplary embodiments of the invention are not limited thereto.

(183) For example, the electron transport region may have an electron transport layer/electron injection layer structure, a hole blocking layer/electron transport layer/electron injection layer structure, an electron control layer/electron transport layer/electron injection layer structure, or a buffer layer/electron transport layer/electron injection layer structure, wherein for each structure, constituting layers are sequentially stacked from the emission layer. However, exemplary embodiments of the structure of the electron transport region are not limited thereto.

(184) The electron transport region (for example, a buffer layer, a hole blocking layer, an electron control layer, or an electron transport layer in the electron transport region) may include a metal-free compound containing at least one π electron-depleted nitrogen-containing ring.

(185) The term “ π electron-depleted nitrogen-containing ring” as used herein refers to a C.sub.1-C.sub.60 heterocyclic group having at least one —N= moiety as a ring-forming moiety.

(186) For example, the “ π electron-depleted nitrogen-containing ring” may be i) a 5-membered to 7-membered heteromonocyclic group having at least one —N= moiety, ii) a heteropolycyclic group in which two or more 5-membered to 7-membered heteromonocyclic groups each having at least one —N= moiety are condensed with each other, or iii) a heteropolycyclic group in which at least one of 5-membered to 7-membered heteromonocyclic groups, each having at least one —N= moiety, is condensed with at least one C.sub.5-C.sub.60 carbocyclic group.

(187) Examples of the π electron-depleted nitrogen-containing ring include an imidazole, a pyrazole, a thiazole, an isothiazole, an oxazole, an isoxazole, a pyridine, a pyrazine, a pyrimidine, a pyridazine, an indazole, a purine, a quinoline, an isoquinoline, a benzoquinoline, a phthalazine, a naphthyridine, a quinoxaline, a quinazoline, a cinnoline, a phenanthridine, an acridine, a phenanthroline, a phenazine, a benzimidazole, an isobenzothiazole, a benzoxazole, an isobenzoxazole, a triazole, a tetrazole, an oxadiazole, a triazine, a thiadiazole, an imidazopyridine, an imidazopyrimidine, and an azacarbazole, but are not limited thereto.

(188) For example, the electron transport region may include a compound represented by Formula 601 below:

[Ar.sub.601].sub.xe11-[(L.sub.601).sub.xe1-R.sub.601].sub.xe21 <Formula 601>

(189) In Formula 601, Ar.sub.601 may be a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, xe11 may be 1, 2, or 3, L.sub.601 may be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted

divalent non-aromatic condensed heteropolycyclic group, xe1 may be an integer from 0 to 5, R.sub.601 may be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.601)(Q.sub.602)(Q.sub.603), —C(=O)(Q.sub.601), —S(=O).sub.2(Q.sub.601), and —P(=O)(Q.sub.601)(Q.sub.602), Q.sub.601 to Q.sub.603 may each independently be a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group, and xe21 may be an integer from 1 to 5.

(190) In one exemplary embodiment, at least one selected from Ar.sub.601(s) in the number of xe11 and R.sub.601(s) in the number of xe21 may include the π electron-depleted nitrogen-containing ring described above.

(191) In one exemplary embodiment, ring Ar.sub.601 in Formula 601 may be selected from: a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, a dibenzothiophene group, a carbazole group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group; and a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, a dibenzothiophene group, a carbazole group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(192) When xe11 in Formula 601 is 2 or more, two or more Ar.sub.601(s) may be linked to each

other via a single bond.

(193) In one or more exemplary embodiments, Ar.sub.601 in Formula 601 may be an anthracene group.

(194) In one or more exemplary embodiments, the compound represented by Formula 601 may be represented by Formula 601-1 below:

(195) ##STR00051##

(196) In Formula 601-1, X.sub.614 may be N or C(R.sub.614), X.sub.615 may be N or C(R.sub.615), X.sub.616 may be N or C(R.sub.616), and at least one selected from X.sub.614 to X.sub.616 may be N, L.sub.611 to L.sub.613 may each be understood by referring to the description provided in connection with L.sub.601, xe611 to may each be understood by referring to the description provided in connection with xe1, R.sub.611 to R.sub.613 may each be understood by referring to the description provided in connection with R.sub.601, and R.sub.614 to R.sub.616 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(197) In one exemplary embodiment, L.sub.601 and L.sub.611 to L.sub.613 in Formulae 601 and 601-1 may each independently be selected from: a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene

group, a tetrazolylene group, an imidazopyridinylene group, an imidazopyrimidinylene group, and an azacarbazolylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group, but exemplary embodiments of the invention are not limited thereto.

(198) In one or more exemplary embodiments, xe1 and xe611 to xe613 in Formulae 601 and 601-1 may each independently be 0, 1, or 2.

(199) In one or more exemplary embodiments, R.sub.601 and R.sub.611 to R.sub.613 in Formulae 601 and 601-1 may each independently be selected from: a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl

group, a phenanthrolyl group, a phenaziny group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenaziny group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; and —S(=O).sub.2(Q.sub.601) and —P(=O)(Q.sub.601) (Q.sub.602), and Q.sub.601 and Q.sub.602 are the same as described above.

(200) The electron transport region may include at least one compound selected from Compounds ET1 to ET36 below, but exemplary embodiments of the invention are not limited thereto:

(201) ##STR00052## ##STR00053## ##STR00054## ##STR00055##

(202) In one or more exemplary embodiments, the electron transport region may include at least one compound selected from 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), Tris-(8-hydroxyquinoline)aluminum (Alq.sub.3), Bis(8-hydroxy-2-methylquinoline)-(4-phenylphenoxy)aluminum (BALq), 3-(biphenyl-4-yl)-5-(4-tert-butylphenyl)-4-phenyl-4H-1,2,4-triazole (TAZ), and 4-naphthalen-1-yl-3,5-diphenyl-1,2,4-triazole (NTAZ):

(203) ##STR00056##

(204) The thickness of the buffer layer, the hole blocking layer, or the electron control layer may be in a range of about 20 Å to about 1,000 Å, for example, about 30 Å to about 300 Å. When the thickness of the buffer layer, the hole blocking layer, or the electron control layer is within the range described above, excellent hole blocking characteristics or electron control characteristics may be obtained without a substantial increase in driving voltage.

(205) The thickness of the electron transport layer may be in a range of about 100 Å to about 1,000 Å, for example, about 150 Å to about 500 Å. When the thickness of the electron transport layer is within the range described above, the electron transport layer may have satisfactory electron transport characteristics without a substantial increase in driving voltage.

(206) The electron transport region (for example, the electron transport layer in the electron transport region) may further include, in addition to the materials described above, a metal-containing material.

(207) The metal-containing material may include at least one selected from an alkali metal complex and an alkaline earth-metal complex. The alkali metal complex may include a metal ion selected from a Li ion, a Na ion, a K ion, a Rb ion, and a Cs ion, and the alkaline earth-metal complex may include a metal ion selected from a Be ion, a Mg ion, a Ca ion, a Sr ion, and a Ba ion. A ligand coordinated with the metal ion of the alkali metal complex or the alkaline earth-metal complex may be selected from a hydroxy quinoline, a hydroxy isoquinoline, a hydroxy benzoquinoline, a hydroxy acridine, a hydroxy phenanthridine, a hydroxy phenyloxazole, a hydroxy phenylthiazole, a

hydroxy diphenyloxadiazole, a hydroxy diphenylthiadiazole, a hydroxy phenylpyridine, a hydroxy phenylbenzimidazole, a hydroxy phenylbenzothiazole, a bipyridine, a phenanthroline, and a cyclopentadiene, but exemplary embodiments of the invention are not limited thereto.

(208) For example, the metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (lithium quinolate, LiQ) or ET-D2 below:

(209) ##STR00057##

(210) The electron transport region may include an electron injection layer that facilitates electron injection from the second electrode **190**. The electron injection layer may be in direct contact with the second electrode **190**.

(211) The electron injection layer may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

(212) The electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof.

(213) The alkali metal may be selected from Li, Na, K, Rb, and Cs. In one exemplary embodiment, the alkali metal may be Li, Na, or Cs. In one or more exemplary embodiments, the alkali metal may be Li or Cs, but exemplary embodiments of the invention are not limited thereto.

(214) The alkaline earth metal may be selected from Mg, Ca, Sr, and Ba.

(215) The rare earth metal may be selected from Sc, Y, Ce, Tb, Yb, and Gd.

(216) The alkali metal compound, the alkaline earth-metal compound, and the rare earth metal compound may be selected from oxides and halides (for example, fluorides, chlorides, bromides, or iodides) of the alkali metal, the alkaline earth-metal, and the rare earth metal.

(217) The alkali metal compound may be selected from alkali metal oxides, such as Li_2O , Cs_2O , or K_2O , and alkali metal halides, such as LiF, NaF, CsF, KF, LiI, NaI, CsI, or KI. In one exemplary embodiment, the alkali metal compound may be selected from LiF, Li_2O , NaF, LiI, NaI, CsI, and KI, but exemplary embodiments of the invention are not limited thereto.

(218) The alkaline earth-metal compound may be selected from alkaline earth-metal oxides, such as BaO, SrO, CaO, $\text{Ba}_x\text{Sr}_{1-x}\text{O}$ ($0 < x < 1$), or $\text{Ba}_x\text{Ca}_{1-x}\text{O}$ ($0 < x < 1$). In one exemplary embodiment, the alkaline earth metal compound may be selected from BaO, SrO, and CaO, but exemplary embodiments of the invention are not limited thereto.

(219) The rare earth metal compound may be selected from YbF_3 , ScF_3 , Sc_2O_3 , Y_2O_3 , Ce_2O_3 , GdF_3 , and TbF_3 . In one exemplary embodiment, the rare earth metal compound may be selected from YbF_3 , ScF_3 , TbF_3 , YbI_3 , ScI_3 , and TbI_3 , but exemplary embodiments of the invention are not limited thereto.

(220) The alkali metal complex, the alkaline earth-metal complex, and the rare earth metal complex may include an ion of alkali metal, alkaline earth-metal, and rare earth metal as described above, and a ligand coordinated with a metal ion of the alkali metal complex, the alkaline earth-metal complex, or the rare earth metal complex may be selected from hydroxy quinoline, hydroxy isoquinoline, hydroxy benzoquinoline, hydroxy acridine, hydroxy phenanthridine, hydroxy phenyloxazole, hydroxy phenylthiazole, hydroxy diphenyloxadiazole, hydroxy diphenylthiadiazole, hydroxy phenylpyridine, hydroxy phenylbenzimidazole, hydroxy phenylbenzothiazole, bipyridine, phenanthroline, and cyclopentadiene, but exemplary embodiments of the invention are not limited thereto.

(221) The electron injection layer may consist of an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combinations thereof, as described above. In one or more exemplary embodiments, the

electron injection layer may further include an organic material. When the electron injection layer further includes an organic material, an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combinations thereof may be homogeneously or non-homogeneously dispersed in a matrix including the organic material.

(222) The thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å, for example, about 3 Å to about 90 Å. When the thickness of the electron injection layer is within the range described above, satisfactory electron injection characteristics may be obtained without a substantial increase in driving voltage.

(223) Second Electrode **190**

(224) The second electrode **190** may be disposed on the organic layer **150** having such a structure. The second electrode **190** may be a cathode which is an electron injection electrode, and in this regard, a material for forming the second electrode **190** may be selected from metal, an alloy, an electrically conductive compound, and a combination thereof, which have a relatively low work function.

(225) The second electrode **190** may include at least one selected from lithium (Li), silver (Ag), magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), ITO, and IZO, but exemplary embodiments of the invention are not limited thereto. The second electrode **190** may be a transmissive electrode, a semi-transmissive electrode, or a reflective electrode.

(226) The second **190** may have a single-layered structure, or a multi-layered structure including two or more layers.

(227) Description of FIGS. **2** to **4**

(228) FIG. **2** is a schematic cross-sectional diagram of another exemplary embodiment of an organic light-emitting device constructed according to principles of the invention. FIG. **3** is a schematic cross-sectional diagram of yet another exemplary embodiment of an organic light-emitting device constructed according to principles of the invention. FIG. **4** is a schematic cross-sectional diagram of still another exemplary embodiment of an organic light-emitting device constructed according to principles of the invention.

(229) An organic light-emitting device **20** of FIG. **2** includes a first capping layer **210**, the first electrode **110**, the organic layer **150**, and the second electrode **190**, which are sequentially stacked in this stated order, an organic light-emitting device **30** of FIG. **3** includes the first electrode **110**, the organic layer **150**, the second electrode **190**, and a second capping layer **220**, which are sequentially stacked in this stated order, an organic light-emitting device **40** of FIG. **4** includes the first capping layer **210**, the first electrode **110**, the organic layer **150**, the second electrode **190**, and the second capping layer **220**, which are sequentially stacked in this stated order.

(230) Regarding FIGS. **2** to **4**, the first electrode **110**, the organic layer **150**, and the second electrode **190** may be understood by referring to the description presented in connection with FIG. **1**.

(231) In the organic layer **150** of each of the organic light-emitting devices **20** and **40**, light generated in an emission layer may pass through the first electrode **110** and the first capping layer **210** toward the outside, wherein the first electrode **110** may be a semi-transmissive electrode or a transmissive electrode.

(232) The first capping layer **210** and the second capping layer **220** may increase external luminescent efficiency according to the principle of constructive interference.

(233) The first capping layer **210** and the second capping layer **220** may each independently be an organic capping layer including an organic material, an inorganic capping layer including an inorganic material, or a composite capping layer including an organic material and an inorganic material.

(234) At least one of the first capping layer **210** and the second capping layer **220** may each independently include at least one material selected from carbocyclic compounds, heterocyclic compounds, amine-based compounds, porphyrine derivatives, phthalocyanine derivatives, a naphthalocyanine derivatives, alkali metal complexes, and alkaline earth-based complexes. The carbocyclic compound, the heterocyclic compound, and the amine-based compound may be optionally substituted with a substituent containing at least one element selected from O, N, S, Se, Si, F, Cl, Br, and I.

(235) In one exemplary embodiment, at least one of the first capping layer **210** and the second capping layer **220** may each independently include an amine-based compound.

(236) In one exemplary embodiment, at least one of the first capping layer **210** and the second capping layer **220** may each independently include the compound represented by Formula 201 or the compound represented by Formula 202.

(237) In one or more exemplary embodiments, at least one of the first capping layer **210** and the second capping layer **220** may each independently include a compound selected from Compounds HT28 to HT33 and Compounds CP1 to CP5, but exemplary embodiments of the invention are not limited thereto.

(238) ##STR00058## ##STR00059##

(239) Hereinbefore, the organic light-emitting device according to an embodiment has been described in connection with FIGS. **1** to **4**, but exemplary embodiments of the invention are not limited thereto.

(240) Layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region may be formed in a certain region by using one or more suitable methods selected from vacuum deposition, spin coating, casting, Langmuir-Blodgett (LB) deposition, ink-jet printing, laser-printing, and laser-induced thermal imaging.

(241) When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by vacuum deposition, the deposition may be performed at a deposition temperature of about 100° C. to about 500° C., a vacuum degree of about 10.sup.-8 torr to about 10.sup.-3 torr, and a deposition speed of about 0.01 Å/sec to about 100 Å/sec by taking into account a material to be included in a layer to be formed, and the structure of a layer to be formed.

(242) When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by spin coating, the spin coating may be performed at a coating speed of about 2,000 rpm to about 5,000 rpm and at a heat treatment temperature of about 80° C. to 200° C. by taking into account a material to be included in a layer to be formed, and the structure of a layer to be formed.

(243) FIG. **5**—Apparatus

(244) FIG. **5** is a schematic diagram of an exemplary embodiment of an apparatus containing an organic light-emitting device constructed according to principles of the invention.

(245) The organic light-emitting device may be included in various apparatuses.

(246) Thus, another aspect of the invention provides an apparatus **300** including the organic light-emitting device **10**. Although the organic light-emitting device **10** is depicted in FIG. **5**, the one or more of the organic light-emitting devices **20**, **30**, and/or **40** may be used instead of or in addition to the organic light-emitting device **10**.

(247) The apparatus **300** may further include, in addition to the organic light-emitting device **10**, a thin film transistor **310**. Here, the thin film transistor **310** may include a source electrode **320**, an activation layer **330**, and a drain electrode **340**, wherein the first electrode **110** of the organic light-emitting device **10** may be in electrical connection with one or both of the source electrode **320** and the drain electrode **340** of the thin-film transistor **310**.

(248) The apparatus **300** may be, for example, a light-emitting apparatus, an authentication apparatus, or an electronic apparatus, but exemplary embodiments of the invention are not limited

thereto.

(249) The light-emitting apparatus **300** may be used as various displays, light sources, and the like.

(250) The authentication apparatus may be, for example, a biometric authentication apparatus for authenticating an individual by using biometric information of a biometric body (for example, a finger tip, a pupil, or the like).

(251) The authentication apparatus may further include, in addition to the organic light-emitting device, a biometric information collector.

(252) The electronic apparatus may be applied to personal computers (for example, a mobile personal computer), mobile phones, digital cameras, electronic organizers, electronic dictionaries, electronic game machines, medical instruments (for example, electronic thermometers, sphygmomanometers, blood glucose meters, pulse measurement devices, pulse wave measurement devices, electrocardiogram (ECG) displays, ultrasonic diagnostic devices, or endoscope displays), fish finders, various measuring instruments, meters (for example, meters for a vehicle, an aircraft, and a vessel), projectors, and the like, but exemplary embodiments of the invention are not limited thereto.

(253) General Definition of Substituents

(254) The term “C.sub.1-C.sub.60 alkyl group” as used herein refers to a linear or branched aliphatic saturated hydrocarbon monovalent group having 1 to 60 carbon atoms, and examples thereof include a methyl group, an ethyl group, a propyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, a pentyl group, an isoamyl group, and a hexyl group. The term “C.sub.1-C.sub.60 alkylene group” as used herein refers to a divalent group having a structure corresponding to the C.sub.1-C.sub.60 alkyl group.

(255) The term “C.sub.2-C.sub.60 alkenyl group” as used herein refers to a hydrocarbon group having at least one carbon-carbon double bond in the middle or at the terminus of the C.sub.2-C.sub.60 alkyl group, and examples thereof include an ethenyl group, a propenyl group, and a butenyl group. The term “C.sub.2-C.sub.60 alkenylene group” as used herein refers to a divalent group having a structure corresponding to the C.sub.2-C.sub.60 alkenyl group.

(256) The term “C.sub.2-C.sub.60 alkynyl group” as used herein refers to a hydrocarbon group having at least one carbon-carbon triple bond in the middle or at the terminus of the C.sub.2-C.sub.60 alkyl group, and examples thereof include an ethynyl group, and a propynyl group. The term “C.sub.2-C.sub.60 alkynylene group” as used herein refers to a divalent group having a structure corresponding to the C.sub.2-C.sub.60 alkynyl group.

(257) The term “C.sub.1-C.sub.60 alkoxy group” as used herein refers to a monovalent group represented by —OA.sub.101 (wherein A.sub.101 is the C.sub.1-C.sub.60 alkyl group), and examples thereof include a methoxy group, an ethoxy group, and an isopropoxy group.

(258) The term “C.sub.3-C.sub.10 cycloalkyl group” as used herein refers to a monovalent saturated hydrocarbon monocyclic group having 3 to 10 carbon atoms, and examples thereof include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, and a cycloheptyl group. The term “C.sub.3-C.sub.10 cycloalkylene group” as used herein refers to a divalent group having a structure corresponding to the C.sub.3-C.sub.10 cycloalkyl group.

(259) The term “C.sub.1-C.sub.10 heterocycloalkyl group” as used herein refers to a monovalent monocyclic group having at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom and 1 to 10 carbon atoms, and examples thereof include a 1,2,3,4-oxatriazolidinyl group, a tetrahydrofuranyl group, and a tetrahydrothiophenyl group. The term “C.sub.1-C.sub.10 heterocycloalkylene group” as used herein refers to a divalent group having a structure corresponding to the C.sub.1-C.sub.10 heterocycloalkyl group.

(260) The term “C.sub.3-C.sub.10 cycloalkenyl group” as used herein refers to a monovalent monocyclic group that has 3 to 10 carbon atoms and at least one carbon-carbon double bond in the ring thereof and no aromaticity, and examples thereof include a cyclopentenyl group, a cyclohexenyl group, and a cycloheptenyl group. The term “C.sub.3-C.sub.10 cycloalkenylene

group” as used herein refers to a divalent group having a structure corresponding to the C.sub.3-C.sub.10 cycloalkenyl group.

(261) The term “C.sub.1-C.sub.10 heterocycloalkenyl group” as used herein refers to a monovalent monocyclic group that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, 1 to 10 carbon atoms, and at least one carbon-carbon double bond in its ring. Non-limiting examples of the C.sub.1-C.sub.10 heterocycloalkenyl group include a 4,5-dihydro-1,2,3,4-oxatriazolyl group, a 2,3-dihydrofuranyl group, and a 2,3-dihydrothiophenyl group. The term “C.sub.1-C.sub.10 heterocycloalkenylene group” as used herein refers to a divalent group having a structure corresponding to the C.sub.1-C.sub.10 heterocycloalkenyl group.

(262) The term “C.sub.6-C.sub.60 aryl group” as used herein refers to a monovalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms, and a C.sub.6-C.sub.60 arylene group used herein refers to a divalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms. Non-limiting examples of the C.sub.6-C.sub.60 aryl group include a phenyl group, a naphthyl group, an anthracenyl group, a phenanthrenyl group, a pyrenyl group, and a chrysenyl group. When the C.sub.6-C.sub.60 aryl group and the C.sub.6-C.sub.60 arylene group each include two or more rings, the rings may be fused to each other.

(263) The term “C.sub.1-C.sub.60 heteroaryl group” as used herein refers to a monovalent group having a carbocyclic aromatic system that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, in addition to 1 to 60 carbon atoms, and the term “C.sub.1-C.sub.60 heteroarylene group” as used herein refers to a divalent group having a carbocyclic aromatic system that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, in addition to 1 to 60 carbon atoms. Non-limiting examples of the C.sub.1-C.sub.60 heteroaryl group include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, and an isoquinolinyl group. When the C.sub.1-C.sub.60 heteroaryl group and the C.sub.1-C.sub.60 heteroarylene group each include two or more rings, the rings may be fused with each other.

(264) The term “C.sub.6-C.sub.60 aryloxy group” as used herein refers to —OA.sub.102 (wherein A.sub.102 is the C.sub.6-C.sub.60 aryl group), and a C.sub.6-C.sub.60 arylthio group as used herein refers to —SA.sub.103 (wherein A.sub.103 is the C.sub.6-C.sub.60 aryl group).

(265) The term “monovalent non-aromatic fused polycyclic group” as used herein refers to a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings fused with each other, only carbon atoms as ring-forming atoms, and no aromaticity in its entire molecular structure. A detailed example of the monovalent non-aromatic fused polycyclic group is a fluorenyl group. The term “divalent non-aromatic fused polycyclic group” as used herein refers to a divalent group having a structure corresponding to the monovalent non-aromatic fused polycyclic group.

(266) The term “monovalent non-aromatic fused heteropolycyclic group” as used herein refers to a monovalent group (for example, having 1 to 60 carbon atoms) having two or more rings fused with each other, at least one heteroatom selected from N, O, Si, P, and S, other than carbon atoms, as a ring-forming atom, and no aromaticity in its entire molecular structure. A detailed example of the monovalent non-aromatic fused heteropolycyclic group is a 1,2,3,4-tetrahydroquinolinyl group. The term “divalent non-aromatic fused heteropolycyclic group” as used herein refers to a divalent group having a structure corresponding to the monovalent non-aromatic fused heteropolycyclic group.

(267) The term “C.sub.5-C.sub.60 carbocyclic group” as used herein refers to a monocyclic or polycyclic group having 5 to 60 carbon atoms in which a ring-forming atom is a carbon atom only. The term “C.sub.5-C.sub.60 carbocyclic group” as used herein refers to an aromatic carbocyclic group or a non-aromatic carbocyclic group. The C.sub.5-C.sub.60 carbocyclic group may be a ring, such as benzene, a monovalent group, such as a phenyl group, or a divalent group, such as a phenylene group. In one or more exemplary embodiments, depending on the number of substituents connected to the C.sub.5-C.sub.60 carbocyclic group, the C.sub.5-C.sub.60

carbocyclic group may be a trivalent group or a quadrivalent group.

(268) The term “C.sub.1-C.sub.60 heterocyclic group” as used herein refers to a group having a structure corresponding to the C.sub.5-C.sub.60 carbocyclic group, except that as a ring-forming atom, at least one heteroatom selected from N, O, Si, P, and S is used in addition to carbon (the number of carbon atoms may be in a range of 1 to 60).

(269) The terms “hydrogen” and “deuterium” refers to their respective atoms and corresponding radicals.

(270) In the exemplary embodiments, at least one substituent of the substituted C.sub.5-C.sub.60 carbocyclic group, the substituted C.sub.1-C.sub.60 heterocyclic group, the substituted C.sub.3-C.sub.10 cycloalkylene group, the substituted C.sub.1-C.sub.10 heterocycloalkylene group, the substituted C.sub.3-C.sub.10 cycloalkenylene group, the substituted C.sub.1-C.sub.10 heterocycloalkenylene group, the substituted C.sub.6-C.sub.60 arylene group, the substituted C.sub.1-C.sub.60 heteroarylene group, the substituted divalent non-aromatic fused polycyclic group, the substituted divalent non-aromatic fused heteropolycyclic group, the substituted C.sub.1-C.sub.60 alkyl group, the substituted C.sub.2-C.sub.60 alkenyl group, the substituted C.sub.2-C.sub.60 alkynyl group, the substituted C.sub.1-C.sub.60 alkoxy group, the substituted C.sub.3-C.sub.10 cycloalkyl group, the substituted C.sub.1-C.sub.10 heterocycloalkyl group, the substituted C.sub.3-C.sub.10 cycloalkenyl group, the substituted C.sub.1-C.sub.10 heterocycloalkenyl group, the substituted C.sub.6-C.sub.60 aryl group, the substituted C.sub.6-C.sub.60 aryloxy group, the substituted C.sub.6-C.sub.60 arylthio group, the substituted C.sub.1-C.sub.60 heteroaryl group, the substituted monovalent non-aromatic fused polycyclic group, and the substituted monovalent non-aromatic fused heteropolycyclic group may be selected from: deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic fused polycyclic group, a monovalent non-aromatic fused heteropolycyclic group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), and —P(=O)(Q.sub.11)(Q.sub.12); a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic fused polycyclic group, and a monovalent non-aromatic fused heteropolycyclic group; a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic fused polycyclic group, and a monovalent non-aromatic fused heteropolycyclic group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic fused polycyclic group, and a monovalent non-aromatic fused heteropolycyclic group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic fused polycyclic group, and a monovalent non-aromatic fused heteropolycyclic group.

fused polycyclic group, a monovalent non-aromatic fused heteropolycyclic group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), and —P(=O)(Q.sub.21)(Q.sub.22); and —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23 and Q.sub.31 to Q.sub.33 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic fused polycyclic group, a monovalent non-aromatic fused heteropolycyclic group, a biphenyl group, and a terphenyl group.

(271) The term “Ph” as used herein refers to a phenyl group, the term “Me” as used herein refers to a methyl group, the term “Et” as used herein refers to an ethyl group, the term “ter-Bu” or “Bu.sup.t” as used herein refers to a tert-butyl group, and the term “OMe” as used herein refers to a methoxy group.

(272) The term “biphenyl group” as used herein refers to “a phenyl group substituted with a phenyl group. In other words, the “biphenyl group” is a substituted phenyl group having a C.sub.6-C.sub.60 aryl group as a substituent.

(273) The term “terphenyl group” as used herein refers to “a biphenyl group substituted with a phenyl group. In other words, the “terphenyl group” is a phenyl group having, as a substituent, a C.sub.6-C.sub.60 aryl group substituted with a C.sub.6-C.sub.60 aryl group.

(274) * and *' as used herein, unless defined otherwise, each refer to a binding site to a neighboring atom in a corresponding formula.

(275) Hereinafter, a compound according to exemplary embodiments and an organic light-emitting device according to exemplary embodiments will be described in detail with reference to Synthesis Examples and Examples. The wording “B was used instead of A” used in describing Synthesis Examples refers to that an identical molar equivalent of B was used in place of A.

EXAMPLES

Example 1

(276) As a substrate and an anode, a glass substrate on which ITO (Corning) 15 Ω /cm.sup.2 (120 nm) was formed was cut to the size of 50 mm×50 mm×0.5 mm, sonicated with acetone, isopropyl alcohol, and pure water, each for 5 minutes, and then, cleaned by exposure to ultraviolet rays and ozone for 10 minutes. Then, the glass substrate was provided to a vacuum deposition apparatus.

(277) Compound HTL1 was vacuum deposited on the ITO anode to form a hole injection layer having the thickness of 600 Å, and 4,4'-bis[N-(1-naphtyl)-N-phenylamino]biphenyl (hereinafter, referred to as NPB) was deposited on the hole injection layer to form a hole transport layer having the thickness of 600 Å.

(278) Compounds HT2-1 and ET2-1 and bis[2-(2-pyridinyl-N)phenyl-C] (acetylacetonato)iridium(III) (Ir(mppy).sub.2(acac)) (8 weight %) which is a phosphorescent dopant were co-deposited on the hole transport layer to form an emission layer having the thickness of 400 Å. Subsequently, Compounds ETL1 and LiQ were mixed (at a ratio of 50:50) and deposited on the emission layer to form an electron transport layer having the thickness of 360 Å, and LiQ was deposited on the electron transport layer to form an electron injection having the thickness of 10 Å. Then, Mg and Ag were co-deposited (at a 10 weight %) on the electron injection layer to form a cathode having the thickness of 130 Å, thereby completing the manufacture of an organic light-emitting device.

(279) ##STR00060##

Examples 2 to 16 and Comparative Examples 1 to 10

(280) Organic light-emitting devices were manufactured in the same manner as in Example 1, except that an emission layer was formed by using compounds listed in Tables 1 and 2 were used. Structures of Compounds A, B, ET1-1, ET1-2, ET3-1, and ET3-2 used in Comparative Examples 1 to 10 are as follows:

(281) ##STR00061##

(282) Synthesis methods of the compounds used in the Examples may be recognized by one of ordinary skill in the art with reference to known techniques in the art.

Evaluation Example 1

(283) The efficiency and lifespan of the organic light-emitting devices manufactured according to Examples 1 to 8 and Comparative Examples 1 to 5 were measured by using Keithley SMU 236 and a luminance meter PR650 at 600 nit, and results thereof are shown in Table 1.

(284) In addition, the efficiency and lifespan of the organic light-emitting devices manufactured according to Examples 9 to 16 and Comparative Examples 6 to 10 were measured by using Keithley SMU 236 and a luminance meter PR650 at 600 nit, and results thereof are shown in Table 2.

(285) In Tables 1 and 2, T97 indicates an amount of time to reach a luminance of 97% of the initial luminance.

(286) TABLE-US-00001 TABLE 1 Lifespan Color (T97) Efficiency coordinate (relative First host Second host (cd/A) CIE_x value) Example 1 HT2-1 ET2-1 162 0.250 115% Example 2 HT2-1 ET2-2 161 0.250 120% Example 3 HT2-2 ET2-1 163 0.250 118% Example 4 HT2-2 ET2-2 161 0.250 120% Example 5 HT1-1 ET2-1 162 0.250 110% Example 6 HT1-1 ET2-2 160 0.250 108% Example 7 HT1-2 ET2-1 162 0.250 115% Example 8 HT1-2 ET2-2 161 0.250 111% Comparative Compound Compound 150 0.250 51% Example 1 A B Comparative HT2-1 — 135 0.250 48% Example 2 Comparative — ET2-2 127 0.250 30% Example 3 Comparative HT1-1 ET1-1 157 0.250 100% Example 4 Comparative HT1-2 ET1-2 157 0.250 98% Example 5

(287) TABLE-US-00002 TABLE 2 Lifespan Color (T97) Efficiency coordinate (relative First host Second host (cd/A) CIE_x value) Example 9 HT2-1 ET4-1 165 0.250 111% Example 10 HT2-1 ET4-2 167 0.250 110% Example 11 HT2-2 ET4-1 164 0.250 113% Example 12 HT2-2 ET4-2 167 0.250 109% Example 13 HT1-1 ET4-1 165 0.250 110% Example 14 HT1-1 ET4-2 168 0.250 106% Example 15 HT1-2 ET4-1 167 0.250 108% Example 16 HT1-2 ET4-2 165 0.250 105% Comparative Compound Compound 150 0.250 74% Example 6 A B Comparative HT2-1 — 135 0.250 65% Example 7 Comparative — ET4-1 115 0.250 52% Example 8 Comparative HT1-1 ET3-1 160 0.250 100% Example 9 Comparative HT1-2 ET3-2 162 0.250 98% Example 10

(288) Referring to Tables 1 and 2, the organic light-emitting devices of Examples 1 to 16 had unexpected and surprising results in terms of high efficiency and excellent lifespan characteristics, as compared with those of the organic light-emitting devices of Comparative Examples 1 to 10.

(289) Accordingly, organic light-emitting device constructed according to the principles and exemplary embodiments of the invention have high efficiency and long lifespan.

(290) Although certain exemplary embodiments and implementations have been described herein, other exemplary embodiments and modifications will be apparent from this description.

Accordingly, the inventive concepts are not limited to such exemplary embodiments, but rather to the broader scope of the appended claims and various obvious modifications and equivalent arrangements as would be apparent to a person of ordinary skill in the art.

Claims

1. An organic light-emitting device comprising: a first electrode; a second electrode facing the first electrode; an organic layer disposed between the first electrode and the second electrode and includes an emission layer; and a first compound and a second compound; wherein the first compound comprises a hole transport compound not including an electron transport moiety, and the

second compound comprises an electron transport compound including an electron transport moiety or a bipolar compound including an electron transport moiety, wherein the second compound includes deuterium, wherein the emission layer comprises the first compound and the second compound, and the second compound is a compound of Formulae 302-1 to 302-4: ##STR00062## wherein: X.sub.321 to X.sub.327, independently of one another, are N or C[(L.sub.324).sub.xb24-R.sub.324], Y.sub.31 is O or S, X.sub.328 is N[(L.sub.325).sub.xb25-R.sub.325] or C[(L.sub.325).sub.xb25-R.sub.325][(L.sub.326).sub.xb26-R.sub.326], xb21 to xb26, independently of one another, are 0, 1, 2, 3, 4, or 5, L.sub.321 to L.sub.326 are each independently selected from: a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl

group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), R.sub.321 to R.sub.326 are each independently selected from a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group; and a C.sub.1-C.sub.60 alkyl group, a C.sub.3-C.sub.10 cycloalkyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), and —N(Q.sub.31)(Q.sub.32), at least one of R.sub.321 to R.sub.326, independently of one another, comprises a C.sub.3-C.sub.10 cycloalkyl group substituted with deuterium, two or more substituents among R.sub.321 to R.sub.326 are, independently, optionally linked to each other to form a substituted or unsubstituted C.sub.1-

C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, Q.sub.31 to Q.sub.33 are each independently selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

2. The organic light-emitting device of claim 1, wherein the electron transport moiety comprises at least one of a cyano group, a phosphine oxide group, a sulfone oxide group, a sulfonate group, and a Ir electron-depleted nitrogen-containing ring.

3. The organic light-emitting device of claim 1, wherein the first compound is a compound of Formulae 301-1 to 301-5: ##STR00063## wherein, in Formulae 301-1 to 301-5, A.sub.301 to A.sub.304, independently of one another, are a benzene group, a naphthalene group, a phenanthrene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, an indole group, a carbazole group, a benzocarbazole group, a dibenzocarbazole group, a furan group, a benzofuran group, a dibenzofuran group, a naphthofuran group, a benzonaphthofuran group, a dinaphthofuran group, a thiophene group, a benzothiophene group, a dibenzothiophene group, a naphthothiophene group, a benzonaphthothiophene group, or a dinaphthothiophene group, X.sub.301 is O, S, N—[(L.sub.305).sub.xb5-R.sub.305], C[(L.sub.305).sub.xb5-R.sub.305][(L.sub.306).sub.xb6-R.sub.306], or Si[(L.sub.305).sub.xb5-R.sub.305][(L.sub.306).sub.xb6-R.sub.306], X.sub.302, Y.sub.301, and Y.sub.302, independently of one another, are a single bond, O, S, N—[(L.sub.305).sub.xb5-R.sub.305], C[(L.sub.305).sub.xb5-R.sub.305][(L.sub.306).sub.xb6-R.sub.306], Si[(L.sub.305).sub.xb5-R.sub.305][(L.sub.306).sub.xb6-R.sub.306], or S(=O).sub.2, xb11 is 1, 2, or 3, xb22 and xb23, independently of one another, are 0, 1, or 2, xb21 and xb24, independently of one another, are 1, 2, 3, 4, or 5, L.sub.301 to L.sub.307, independently of one another, are a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic fused polycyclic group, or a substituted or unsubstituted divalent non-aromatic fused heteropolycyclic group, xb1 to xb6, independently of one another, are 0, 1, 2, 3, 4, or 5, xb7 is 1, 2, 3, 4, or 5, R.sub.301 to R.sub.306, independently of one another, are deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic fused polycyclic group, a substituted or unsubstituted monovalent non-aromatic fused heteropolycyclic group, —Si(Q.sub.301)(Q.sub.302)(Q.sub.303), —N(Q.sub.301)(Q.sub.302), —B(Q.sub.301)(Q.sub.302), —C(=O)(Q.sub.301), —S(=O).sub.2(Q.sub.301), or —P(=O)(Q.sub.301)(Q.sub.302), R.sub.311 to R.sub.314, independently of one another, are hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group —Si(Q.sub.301)(Q.sub.302)(Q.sub.303), —N(Q.sub.301)(Q.sub.302), —B(Q.sub.301)(Q.sub.302), —C(=O)(Q.sub.301), —S(=O).sub.2(Q.sub.301), or —

P(=O)(Q.sub.301)(Q.sub.302), and Q.sub.301 to Q.sub.303, independently of one another, are hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic fused polycyclic group, a monovalent non-aromatic fused heteropolycyclic group, a biphenyl group, or a terphenyl group.

4. The organic light-emitting device of claim 1, wherein the first compound is a fluorene-containing compound, a carbazole-containing compound, a diarylamine compound, a triarylamine compound, a dibenzofuran-containing compound, a dibenzothiophene-containing compound, or a dibenzosilole-containing compound.

5. The organic light-emitting device of claim 1, wherein the first compound comprises deuterium.

6. The organic light-emitting device of claim 3, wherein R.sub.301 to R.sub.306, and R.sub.311 to R.sub.314, independently of one another, are deuterium, a C.sub.1-C.sub.60 alkyl group substituted with deuterium, a C.sub.3-C.sub.10 cycloalkyl group substituted with deuterium, a C.sub.6-C.sub.60 aryl group substituted with deuterium, or a C.sub.1-C.sub.60 heteroaryl group substituted with deuterium.

7. The organic light-emitting device of claim 1, wherein the second compound is a pyridine-containing compound, a pyrimidine-containing compound, a triazine-containing compound, a triazole-containing compound, an imidazole-containing compound, or an oxazine-containing compound.

8. The organic light-emitting device of claim 1, wherein the first compound is a compound of HT1-1, HT1-2, HT2-1, or HT2-2: ##STR00064## ##STR00065##

9. An organic light-emitting device comprising: a first electrode; a second electrode facing the first electrode; an organic layer disposed between the first electrode and the second electrode and includes an emission layer; and a first compound and a second compound; wherein the first compound comprises a hole transport compound not including an electron transport moiety, and the second compound comprises an electron transport compound including an electron transport moiety or a bipolar compound including an electron transport moiety, wherein the second compound includes deuterium, wherein the emission layer comprises the first compound and the second compound, and wherein the second compound is a compound of ET4-1 or ET4-2: ##STR00066##

10. The organic light-emitting device of claim 1, wherein the emission layer comprises a first emission layer and a second emission layer, the first emission layer comprises a first host and a second host, and the second emission layer comprises a third host and a fourth host, wherein the first host in the first emission layer and the third host in the second emission layer each include the first compound, and the first host and the third host are same or different from each other, the second host in the first emission layer and the fourth host in the second emission layer each include the second compound, the second compound included in the second host comprises the electron transport compound, the second compound included in the fourth host comprises the bipolar compound, the first host and the second host each satisfy Equation 1 below, and the third host and the fourth host each satisfy Equation 2 below:

$E_{\text{sub.HOMO1}} - E_{\text{sub.HOMO2}} \leq 0.2 \text{ eV}$ <Equation 1>

$E_{\text{sub.HOMO3}} - E_{\text{sub.HOMO4}} \geq 0.3 \text{ eV}$ <Equation 2> wherein, in Equations 1 and 2,

$E_{\text{sub.HOMO1}}$ indicates a highest occupied molecular orbital (“HOMO”) energy level of the first host, $E_{\text{sub.HOMO2}}$ indicates a HOMO energy level of the second host, $E_{\text{sub.HOMO3}}$ indicates a HOMO energy level of the third host, and $E_{\text{sub.HOMO4}}$ indicates a HOMO energy level of the

fourth host.

11. The organic light-emitting device of claim 1, wherein the emission layer emits green light having a maximum emission wavelength in a range of about 490 nm to about 560 nm.
 12. The organic light-emitting device of claim 1, wherein the emission layer further comprises a dopant, and the dopant comprises a phosphorescent dopant or a fluorescent dopant.
 13. The organic light-emitting device of claim 1, wherein: the first electrode is an anode, the second electrode is a cathode, the organic layer, further comprises at least one of a hole transport region disposed between the first electrode and the emission layer and at least one of an electron transport region disposed between the emission layer and the second electrode, the hole transport region comprises a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and the electron transport region comprises a hole blocking layer, an electron transport layer, an electron injection layer, or any combination thereof.
 14. The organic light-emitting device of claim 13, wherein the hole transport region comprises an amine-based compound.
 15. The organic light-emitting device of claim 13, wherein the electron transport region comprises a metal-containing material.
 16. An apparatus comprising: the organic light-emitting device of claim 1, wherein the first electrode of the organic light-emitting device is in electrical contact with one of a source electrode and a drain electrode of a thin-film transistor.
-